DAY 07 | POST-READ

Relationships and Docs: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Constraints	primary_key, unique, nullable, default, ForeignKey. Enforced by the database.
One-to-many	ForeignKey on the many side, relationship() with back_populates on both.
Many-to-many	Association table of id pairs, relationship(secondary=table).
Navigation	author.books, book.author, book.tags work like normal Python attributes.
Transactions	A unit of work: commit on success, rollback on error.
ACID	Atomic, Consistent, Isolated, Durable.
OpenAPI	Machine-readable API contract at /openapi.json.
Swagger and ReDoc	/docs to try requests, /redoc to read. Both from the same schema.
Better docs	App title, version, description; tags, summary, responses, Field examples.

Part B: Going beyond today

1. The N+1 query problem

Listing 100 books and then touching book.author for each can fire 101 queries. Load related data up front:

```
from sqlalchemy.orm import selectinload
stmt = select(Book).options(selectinload(Book.author), selectinload(Book.tags))
```

2. Cascades

```
books: Mapped[list["Book"]] = relationship(back_populates="author", cascade="all, delete-orph
```

Deleting an author now deletes their books too. Powerful, so use it deliberately.

3. Migrations with Alembic

create_all only creates missing tables; it never changes existing ones. In real projects, Alembic records every schema change as a versioned migration script you can apply or undo.

4. session.begin() context manager

```
with Session(engine) as s, s.begin():
    s.add(order)
    s.add(payment)
# commits automatically, or rolls back if an exception is raised
```

5. Generating clients from OpenAPI

Tools such as openapi-generator can read /openapi.json and produce a ready-made client library for TypeScript, Java or Python. Your docs become code for other teams.

Part C: Check yourself

- On which table does the foreign key go in a one-to-many relationship?
- Why does many-to-many need an extra table?
- What happens to the data if an exception occurs before commit?
- What is the difference between /docs and /openapi.json?
- Name three things that make API docs genuinely useful.

Part D: Practice ideas for tonight

- Model students, courses and enrolments (many-to-many) and list the courses for one student.
- Write a bulk import function that rolls back if any single record is invalid.
- Add descriptions, examples and error responses to every endpoint in your Library API.

Coming up on Day 8: dependency injection, middleware, JWT authentication, and using PyMongo from FastAPI.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today